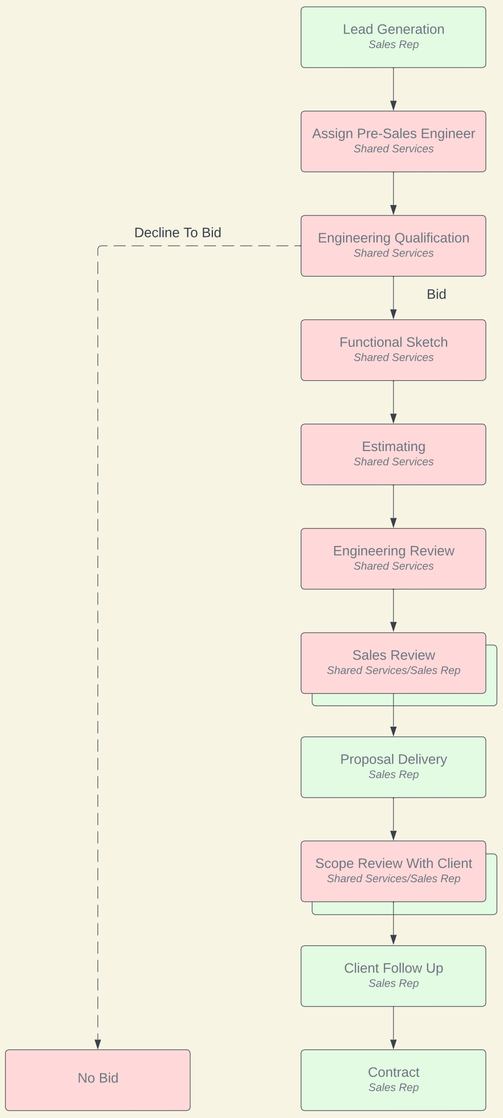
INSTRATA™ AV PROCESS FLOW CHART

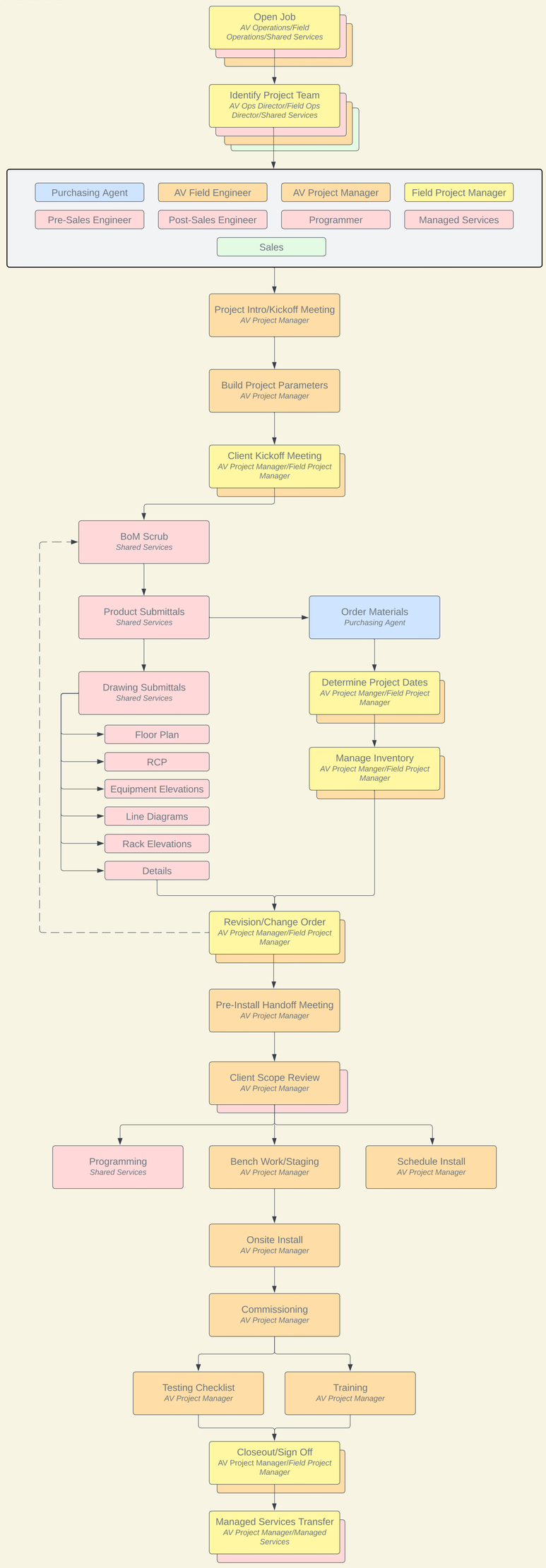
Diagram Key

- Sales Rep
- Shared Services
- Field Operations
- Purchasing
- AV Operations

Pre-Sales



Post-Sales



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Pre-Sales Process Details

I. Lead Generation

a. Owner: Sales

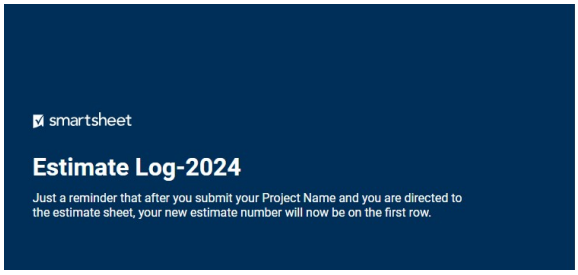
b. Action Item: Add lead to estimate log/Monday/email

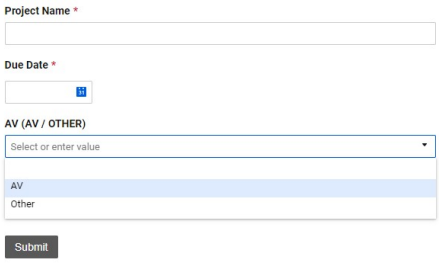
c. Summary

i. Lead generation may happen via several methods. Ultimately the passing of leads from the sales/estimating team to the shared services team is the role of the sales rep.

d. Details

i. Bid Log Entry: Each individual business unit may add an entry into their bid log (smartsheets) and as long as the entry is marked as an AV project (see image below) the entry will also be copied into Monday (AV workflow system) as a new task.





ii. Email: Any sales rep can also email leads directly to Seth Powell (seth.powell@kane.com) and Lee Thompson (lee.thompson@kane.com) to add into our system. If using the bid log, please also send an email with any pertinent information regarding the project (Floor plans, full RFP documents/drawing sets, functional scopes, product specifications, client information, budget information, etc.).

iii. Monday: By clicking this link, you can access our entry form for Monday which allows you to add a project directly to Monday bypassing other methods. We would prefer this be used as a last effort if the other methods aren't accessible as entering here and the bid log will cause duplicate entries in our system. Project files can be uploaded directly to this form as well.

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e. Notes

i. Please be as detailed as possible with job information as this will help our team in identifying project scope and getting going.

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II. Assign Pre-Sales Engineer

a. Owner: AVSS

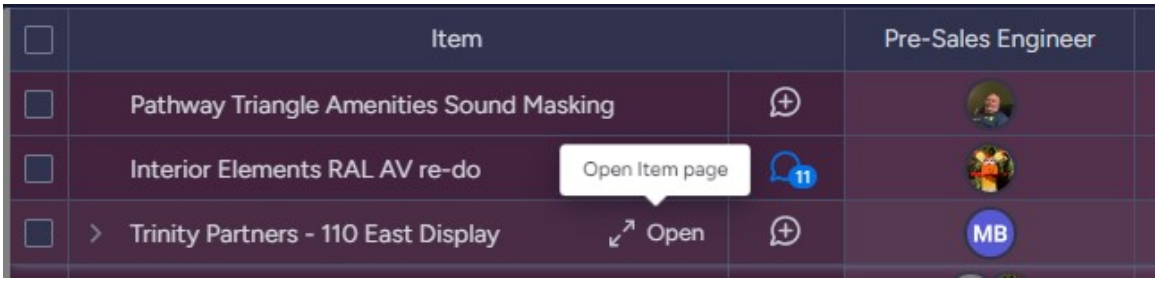
b. Action Item: Assign a Pre-Sales Engineer in Monday

c. Summary

i. Once a lead has been entered into our system, we will assign a pre-sales engineer based on the following criteria (and in this order): Location, Current Workload, Turnaround Time.

d. Details

i. Once a pre-sales engineer has been assigned on our side, the pre-sales engineer will make contact with the sales rep/estimator assigned to the project within 24 hours to begin conversations about the project and to get started with our engineering process. The sales rep should expect to be ready to review all project documentation as well as schedule/setup a discovery call with the client if it is necessary. Lastly, the sales rep should define a preferred sales review meeting date/time. This due date would be different than the project due date (determined by the actual bid due date/time). The sales review date would be the date that the sales person expects us to be done with our package so they can have time to format in any way they would prefer to deliver to the customer.



e. Notes

i. Please keep in mind that the more time we have to quote a project the more detailed we can be. We prefer to have two weeks time to put a proposal together. We recognize that this doesn't always happen. If you're dealing with a short timeline, please try your best to be available for a scope review the day before or day of

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submittal for the sales review so as to maximize engineering time on a project.

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III. Engineering Qualification

a. Owner: AVSS

b. Action Item: Review project within 24 hours

c. Summary

i. Once all information is gathered for a specific project (see Assign Pre-Sales Engineer step above) the pre-sales engineer will then look through all project documentation to ensure that this is a project that shared services is capable of completing. If the project meets our criteria, we will move forward with quoting it. If it does not, the project will be no-bid.

d. Details

i. There are several criteria that we will take into account when doing our engineering qualification. One is that all necessary information is present in the package. This will vary by package and bid type. Some bids will have an RFI period where we can ask any questions that we need answered in order to complete our proposal. Others may not have this option or may be presented as a bid but with no real product specification. Other criteria we will be looking at is delivery time (we would prefer to have a minimum of 2 weeks but can be flexible as our workload allows), viability of the project (is this a bid to a client we've worked with before or is this just a shot in the dark), and overall complexity of the project and our ability to perform it (as an example if the entire project is specified with products we've never seen or heard of we would likely opt to no-bid).

e. Notes

i. This engineering qualification is a two way conversation and is not intended to allow us to no-bid without a discussion with the sales team. If we decide this should be a no-bid on the engineering side, we will communicate this along with our reasoning to the sales team involved and discuss our next steps. If this is something the sales team needs us to bid for one reason or another, we're happy to accommodate these requests.

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IV. Functional Sketch

a. **Owner: AVSS**

b. **Action Item:** Line diagram sketch per room designed

c. **Summary**

   i. The shared services team will begin our process by documenting system flow with a simple line diagram for each room. This may not be necessary if the bid specifications already show signal flow diagrams. If they do not, this step allows us to clearly demonstrate system functionality across all team members quickly and helps us to not miss parts as we quote the project.

d. **Details**

   i. These drawings will either be done by hand on a piece of paper and uploaded to the project folder (AV Shared Services), via Visio or as a smartboard or whiteboard sketch.

e. **Notes**

   i. This is an INTERNAL ONLY document. This document should never be shown to the client as it is for internal documentation/completion purposes. These drawings will not have title blocks and will likely not be in a presentable form (hand drawn) so the intention is to never show these drawings to the client.

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V. Estimating

a. Owner: AVSS

b. Action Item: Full DTools Proposal with Project Summary

c. Summary

i. The shared services team will create a project within DTools (our quoting platform). The project will either have a location identified per room or per room type (depending on how the project is broken down). This will equate to how the quote itemizes the systems. For example, we could name a location Small Conference (Typical of 12) and just show QTY 12 (or whatever is required) of every system in that bundle or we could name a location Small Conference 201 and just show QTY 1 (or whatever is required) of the single room stated in the location.

d. Details

i. Proposals will be delivered in DTools report format (a pdf file with a specific look/feel) which is customized by business unit with the logo for the corresponding business unit as well as language which reflects the title of each individual business unit (for example, terms and conditions for NET100 all say “NET100 is responsible for…” etc. If another format is desired (as an example, the standard pdf template that other branches use for bidding) we can provide any details needed to add to an existing format. We can provide project information including bill of materials by room type, room pricing breakouts, detailed scope of work in rich text format, and job gross margin information among other things. Please specify early what format you believe is needed for this proposal in order for us to prepare all documents before our review.

e. Notes

i. If any special pricing or specific items need to be added to our proposal, please plan to review those with our shared services team ahead of time.

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Cafe

8	**Biamp Systems**	Two-Way 6.5-Inch Full Range High Output Ceiling Loudspeaker, 8 Ohm
3	**Chief**	ADJ. PIPE 84" TO 108
3	**Chief**	SINGLE CEILING MOUNT LARGE BLACK
3	**Extron**	DTP HDMI 4K 230 Rx - HDMI Twisted Pair Receiver - 230 feet (70 m)
3	**Samsung**	65-inch Commercial 4K UHD Display, 350 NIT
1	**Premiere**	Miscellaneous Cables, Connectors and Hardware
1	**Premiere**	Shipping and Handling

Equipment:	**$10,043.39**
Labor:	**$3,885.98**
Cafe Total	**$13,929.37**

PROJECT SUMMARY

Equipment:	$127,458.97
Labor:	$37,669.70
Premiere Protect:	$22,942.61
Sales Tax:	$13,635.17
Grand Total:	**$201,706.45**

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VI. Engineering Review

a. **Owner: AVSS**

b. **Action Item:** Review proposal/systems with Engineering Team

c. **Summary**

i. Before a quote is finalized on our side, a review meeting is scheduled with Seth Powell or Lee Thompson to cover all pertinent job details and quote details.

d. **Details**

i. In this review meeting we are checking to make sure labor rates are correct, all items are included per the project documents and miscellaneous items like taxes and shipping and handling are all accounted for. Typically this is a 30min to 1hr meeting to review all project specifics in detail before the sales review meeting.

e. **Notes**

i. Due to the technical nature of this call and the unique systems we are using to quote projects specific to AV, this review meeting will be with a shared services director and pre-sales engineer exclusively.

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VII. Sales Review

a. Owner: Sales

b. Action Item: Engineer to review quote with Sales to dial in sell price (labor rates, taxes, local market adjustments, etc).

c. Summary

i. Once a project has been approved on the engineering side, a sales review meeting will be setup with the sales person/estimator and anyone else from the business unit that wants to be involved. The nature of this call is a high level sales overview of what we've provided in our proposal as well as putting any finishing touches on pricing.

d. Details

i. Specific pricing items we would like to review as part of this call is labor cost and sale rates, non-productive labor rate (by default this is 3% of overall project hours but can be modified to fit the needs of the project) and tax rates. Other items that we can review is a brief room summary defining which rooms are included in our project, a rundown of miscellaneous labor items like engineering time, travel, and equipment rental (among others) and we can also discuss delivery format again.

e. Notes

i. This meeting should be scheduled at the beginning of the pre-sales process when the pre-sales engineer first makes contact with the salesperson. This ensures that we have a target date to review the project and no one is rushing once the project is due to get the quote delivered to the customer.

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VIII. Proposal Delivery

a. Owner: Sales

b. Action Item: Send proposal to customer/Schedule a scope review meeting

c. Summary

i. Once the proposal has been reviewed by both the shared services team and the sales team and approved for delivery, the proposal will be delivered specifically by the salesperson or estimator.

d. Details

i. If the project is a “House” project that doesn’t have an assigned salesperson or estimator, it is acceptable for the pre-sales engineer to deliver the project to the customer themselves. However, this should always be a last resort. If the salesperson/estimator would like to copy the pre-sales engineer on the quote delivery email, that is also acceptable for any follow up that may be necessary.

e. Notes

i. We have had a lot of luck with requesting client scope review meetings at time of proposal delivery. Our chances of winning a project will always be higher when we have a chance to explain our design process and give the customer a good rundown of what we’ve specified and why. It makes our team look confident and ready to execute once they sign the contract.

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IX. Scope Review With Client

a. **Owner: Sales**

b. **Action Item:** Host a scope review meeting with client to go over what we've quoted. Include the pre-sales engineer on this call.

c. **Summary**

   i. The scope review with the client should always include both the salesperson/estimator and the pre-sales engineer at a minimum. It's important to have both sales and engineering representation on the call to be able to answer any kind of question they may have about the project.

d. **Details**

   i. Generally this would be an hour long remote conference call where anyone acting as a client representative is welcome to present any questions they may have regarding what we've put together. Sometimes these calls are more sales related and sometimes more engineering related depending on the attendees of the meeting and how savvy they are with AV systems. Our team will come prepared to discuss our systems in depth and take notes on any changes that need to be made or revisions that need to be done.

e. **Notes**

   i. The takeaway from this meeting should be any notes regarding the project as well as a timeline for re-bid or updated quotes.

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X. Client Follow Up

a. Owner: Sales

b. Action Item: Weekly follow up. Be ready to discuss progress on bi-weekly “Jobs in Progress” meetings.

c. Summary

i. The client follow up is the sole responsibility of the salesperson/estimator involved unless this is a House account in which case the pre-sales engineer will be the point of follow up.

d. Details

i. In general, a weekly update of where the project stands would be sufficient for our reporting systems. The general idea of what we want to know is whether or not the project has been awarded (won/lost) and an updated win percentage if it has not been awarded yet (25%, 50%, 75%). If the project has been lost, please provide the reason the project was lost for tracking purposes.

e. Notes

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XI. Contract

a. **Owner: Sales**

b. **Action Item:** Open project at the local business unit. Inform the AVSS team that the project has been won and communicate the job number for our records.

c. **Summary**

   i. Once a contract has been awarded, please notify the shared services team immediately so we can begin our post-sales cycle.

d. **Details**

   i. The most important information we need on this step is just a win date and a signed contract.

e. **Notes**

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Post-Sales Process Details

I. Open Job

a. Owner: Operations

b. Action Item: Open a job at the local business unit. Communicate job number to AVSS.

c. Summary

i. Once a project has been awarded, our team begins the post-sales cycle by working with the individual business unit to get a job opened or created for the project.

d. Details

i. Our team can collaborate with each individual business unit to get all pertinent job information over for opening the project on the post-sales side. This includes but is not limited to labor cost, labor price, equipment cost, equipment price, job gross margin, taxes, and other pertinent information. The salesperson/estimator should mark the job as won in the smartsheet bid log.

e. Notes

i. We have a detailed open job report we can print in DTools which should have all pertinent job information included in it. This will be sent upon request and can be modified as needed to add additional information. Our goal is to streamline the job opening process to make it as easy as possible on all teams.

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II. Identify Project Team

a. **Owner: Operations/AV Ops Manager**

b. **Action Item:** Create an MS Teams group for the project including all project team members.

c. **Summary**

i. Once a job has been opened, the AV Ops Manager, Field Ops Director and Shared Services team will then assign a project team. This project team includes the purchasing agent, AV field engineer, AV project manager, field project manager, pre-sales engineer, post-sales engineer, programmer (if needed – based on programming or DSP needs), managed services team (if needed – based on premiere protect) and Salesperson/estimator.

d. **Details**

i. The AV Ops Manager will create a teams chat that is project specific with all project team members included. The idea behind this step is to create a chat where all project information can go and be discussed so as to facilitate better communication throughout the lifecycle of the project. Any questions related to current jobs can be sent via email to the necessary parties but should also be posted in the teams chat so that everyone has visibility of what is happening behind the scenes at any given point.



e. **Notes**

i. Please note that each person assigned to the project will play a specific role and not everyone will be able to answer every question.

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III. Project Intro/Kickoff Meeting

a. **Owner: AV Project Manager**

b. **Action Item:** Schedule and host AV Kickoff meeting

c. **Summary**

   i. This will be a 30 minute teams meeting with the entire project team in which the pre-sales engineer will give a short project overview to bring everyone else up to speed on what the project will consist of and any other information pertinent to the project at this point.

d. **Details**

   i. The AV project manager will be the one responsible for coordinating and scheduling this meeting. The meeting should be scheduled as a Microsoft teams call so that anyone working remote can join as well. No further steps will be taken on the post-sales side until this meeting has happened as it is the first step in bringing everyone up to speed on the operations side of the project.

e. **Notes**

   i. The pre-sales engineer should come prepared to discuss this project at a high level (end client details, basic functionality overview, any timelines they are privy to, etc).

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IV. Build Project Parameters

a. Owner: AV Project Manager

b. Action Item: Gather project information

c. Summary

i. This will be the first iteration of information gathering by the operations team. Important information to gather would be the following:

1. End clients current construction schedule
2. TCO date (or date they plan to have all construction completed and site clean and ready for AV install)
3. Any information about other trades we may need to coordinate with (furniture vendor, electrician, etc)
4. Dates that all deliverables would be needed (specifically equipment elevations for displays, product submittals and install drawing sets)

d. Details

i. A basic overview of information in this step should be enough to understand what we need to do. Ultimately the goal of this step is to gain some insight into the customers timeline expectations and see how that matches our stated number of install hours and programming hours. Essentially, will our quoted timeline meet the customers current expectations or will there be a gap in that scheduling that we need to discuss (i.e. if the customer has a 3 week AV install timeline budgeted but we have 5 weeks of time on the project).

e. Notes

i. Any documents that are uncovered during this process should be shared with the AV team in the teams group chat or files related to the team associated with this project in teams.

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V. Client Kickoff Meeting

a. **Owner: Operations**

b. **Action Item:** Schedule and host meeting with client to relay expectations of project timeline on our side.

c. **Summary**

i. In this step, the AV Project Manager and the Field Project Manager will get together with the client to introduce themselves as the project leads as well as discuss any expectations that need to be set regarding timelines from the previous step.

d. **Details**

i. The most important aspect of this step is making sure the client understands the timeline we will be working on and ensuring if any scheduling changes need to occur to accommodate us that those are beginning to be discussed here rather than later in the project and showing up as a surprise to the end client. If our schedule is not going to be sufficient for the end client, we should be discussing the idea of a change order to accommodate (whether that means overtime or more technicians being allocated to the project).

e. **Notes**

i. Any takeaway notes from this meeting should be put into the teams chat for this AV project so that the whole project team can see what was discussed and work on any action items (like a change order for updated labor).

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VI. BoM Scrub

a. **Owner: AVSS**

b. **Action Item:** Review BoM to make sure all items are accounted for.

c. **Summary**

i. In this step the bill of materials included as part of the scope will be reviewed by the engineering and AV operations group for completeness.

d. **Details**

i. This will be a short and simple engineering review. All miscellaneous cables connectors and hardware WILL NOT be included in this initial bill of materials review as these will be completed and turned in as part of the AV drawing package. This list will comprise of short interconnect cables, specialty adapters and any additional equipment that was missed as part of the original AV design. The goal on the engineering side is to capture everything except for short interconnect cables as part of the original bill of materials.

e. **Notes**

i. If this project is a design/build project and no product submittals are required, once this step is complete the bill of materials can be sent to purchasing for order. If this is a bid or if product submittals are required by the GC, please move to the next step in the process. Please note: some projects may have a phased purchasing plan depending on project schedule and product availability. The purchasing agent involved in the project will work directly with operations to define how to purchase the BoM in a timeline and efficient manner.

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VII. Product Submittals

a. **Owner: AVSS**

b. **Action Item:** Create a full set of project submittals combined to a PDF with a table of contents noting pages of each device in the BoM.

c. **Summary**

   i. If product submittals are required (usually required for a bid scenario or if the GC is requesting submittals) the shared services team will create a product submittals doc within DTools.

d. **Details**

   i. The format of the DTools submittals document is that all products are listed per room with links to the individual cut sheets, data sheets and install manuals (where applicable). Along with this document, we will also submit a combined set of all PDF documents in that set as a separate document (for printing and delivering to the end client). If a specific format is required for submittal documentation, please let shared services know this up front so that we can create a plan for getting the submittals into the proposed format.

e. **Notes**

   i. Returned submittals are usually sent back one of three ways:

      1. Revise and Resubmit: this means there are problems with our submittals that are usually stated as comments in the original set. If this is what we receive, please send this document to the post-sales engineer ASAP so we can revise and return the submittals with the stated changes

      2. Approved as Noted: this means that they've made some comments (usually pertaining to product color or other options) and means we do not need to resubmit any submittals documents as long as we can abide by the comments they've added. Please communicate this document with the whole team so we can make any small changes necessary to the bill of materials before ordering.

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3. Approved: If the submittals are approved without any notes, the bill of material can be released for order. The purchasing agent and operations team will determine the best course of action for ordering equipment based on project schedule and materials availability.

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Product Files

Large Conference Room



1 **Newline TT-7523NT-PLUS**
75" NT+ Series Commercial Display

Files
Spec Sheet
Newline_DataSpec Sheet_emYYn4sR30uYixkm8Fg_WQ.pdf



1 **Chief XTM1U**
Micro-Adjust Tilt Wall Mount X-Large

Files
User Manual
Chief_XTM1U_Manual1.pdf



1 **Extron 60-1563-102**
TLS 725M - 7" Wall Mount TouchLink Scheduling Panel - Black

Files
Spec Sheet
Extron_60-1563-102_Brochure3.pdf
User Manual
Extron_60-1563-102_Manual1.pdf



1 **LOGITECH 960-001308**
Rally Bar - Graphite

Files
Spec Sheet
Logitech_960-001308_DataSpec Sheet1.pdf



1 **LOGITECH 989-000430**
Rally Mic Pod Graphite

Files
Spec Sheet
Logitech_989-000430_Brochure1.pdf

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VIII. Order Materials

a. **Owner: Purchasing**

b. **Action Item:** Order known equipment

c. **Summary**

i. Once submittals are approved (or once the BoM is scrubbed if no submittals are required) a materials list should be sent to the business units local purchasing team by the project manager for ordering.

d. **Details**

i. The purchasing agent will order all materials associated with the project (except for miscellaneous equipment which would be sent shortly after finishing the drawing sets) and they would then enter all tracking/lead time information into a purchasing module for everyone in the project to be able to see. This purchasing module will live both in the Microsoft Teams chat for the project as well as the project folder for the job.

ii. If lead times are an issue on any project, we should be letting the end client know these things as soon as possible and ask if they would like to consider alternative solutions or just wait for the long lead time equipment to come in. If they want an updated quote for a similar product, please loop back in the pre-sales engineer for a new change order. If they prefer to wait on the long lead time item, please make sure all dates are updating accordingly and any schedule changes should be relayed to the customer.

e. **Notes**

i. In general, it is hard to give realistic lead times until an order is placed to the manufacturer. If you are asked by the end client directly for lead times on equipment before we receive the lead times, a general rule of thumb would be to say we can convey real lead times once we have them but 6-8 weeks is pretty standard on most items that are non-custom items.

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IX. Drawing Submittals

a. **Owner: AVSS**

b. **Action Item:** Create all construction drawing submittals

c. **Summary**

i. Immediately following the project kickoff meeting, the post-sales engineer will begin system drawings. The drawings include but are not limited to:

1. Floor Plan
2. Reflected Ceiling Plan
3. Equipment Elevations
4. Line Diagrams
5. Rack Elevations
6. Detail Drawings

d. **Details**

i. Floor plans, equipment elevations and line diagrams will be included in every package. Reflected ceiling plans, rack elevations and detail drawings will be provided if needed (i.e. we won't be providing any reflected ceiling plans if no items are being installed in the ceiling.

ii. As part of our standard line diagram package, we will also include a detailed list of "pre-terminated cables and adapters" which will complete our entire bill of materials for this project. Once the project manager receives the system drawings, they should send the list of miscellaneous cables and adapters to purchasing to complete all orders for this project.

e. **Notes**

i. In an effort to reduce time involved with the project, all drawing submittals will be produced to look and feel exactly the same (numbering scheme, format, cable labels etc). All templates have been pre-defined by the shared services team. If special accommodations need to be made for the drawing set for a particular project (i.e. if they need cables labeled a very specific way) please let the post-sales engineer know ASAP so they can include this information in the original set.

ii. All system drawings are done in CAD format and printed to PDF.

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X. Determine Project Dates

a. **Owner: Operations**

b. **Action Item:** Communicate accurate install dates/durations with project team.

c. **Summary**

   i. While drawing submittals are being processed by the shared services team, the project managers involved on the project should be putting together an SSP for the project which details specific dates that we will be installing as well as a basic plan for how to accomplish the overall scope (order of rooms to be completed, scheduled timeline of events, allocating technician resources to get the job completed in the allocated time).

d. **Details**

   i. Any internal scheduling details should be posted to the teams chat for that particular job so all parties are made aware of any dates. As dates change, please make sure to put all new dates in the chat as well so that the shared services team can prioritize jobs accordingly.

e. **Notes**

   i. It is also important that any install dates be relayed directly to the client and get full approval for our timeline. This step should have been effectively mitigated if we have already relayed our install timeline to the customer in the above steps.

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XI. Manage Inventory

a. **Owner: Operations**

b. **Action Item:** As product arrives, stage it in the warehouse by room and log all serial numbers and MAC addresses.

c. **Summary**

   i. As the equipment arrives at the warehouse, the project managers should be checking in all equipment and placing it in it's dedicated space for the job in the warehouse.

d. **Details**

   i. It is the responsibility of the project managers to ensure all equipment is delivered and staged appropriately. Please keep track of location of all equipment.

   ii. A spreadsheet should be updated with all products that have been delivered along with serial numbers and MAC addresses. This spreadsheet should live on the teams thread for the associated job.

e. **Notes**

   i. Preferably equipment is staged by room. We usually put a sticker on each piece of equipment with the room that it goes in so we can easily palletize by room when we're ready to deliver to the site.

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XII. Revision/Change Order

a. **Owner: Operations**

b. **Action Item:** Relay any change order requests to the AVSS team for processing.

c. **Summary**

i. Inevitably throughout the lifecycle of a project, there could be change orders that are warranted. Anytime a change order is needed, the project manager should involve the pre-sales engineer to re-engage the project and put together any necessary change orders for review. When the pre-sales engineer finishes the CO, it will be reviewed by the salesperson, just like in the normal pre-sales process and then forwarded to the customer for approval. Once the change orders are approved, they go back into the bill of materials scrub process.

d. **Details**

i. Internal Change Orders: If a project needs a change order that is a zero dollar change order (direct flip of parts to something around the same cost) or if we are putting together a change order because of parts that we forgot, the change order should not go to the customer and will be processed as an internal change order.

ii. External Change Orders: An external change order would be needed if the design functionality changes in any way and requires additional labor or equipment.

e. **Notes**

i. Anytime a project is marked as approved in DTools (when a project is won), any additional parts or labor adds will be presented as a change order. These change orders can be marked as either internal (doesn't affect the price to the customer) or external (does affect the price to the customer).

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XIII. Pre-Install Handoff Meeting

a. **Owner: AV Project Manager**

b. **Action Item:** Schedule and host the project handoff meeting with all project team members.

c. **Summary**

i. Once all deliverables are completed, and in theory about 2 weeks before the system is deployed, the AV project manager will schedule a 1 hour remote call with all of the project team to discuss the project in detail.

d. **Details**

i. The purpose of this call is to go through all project details. Specifically we will be reviewing all submittal drawings and discussing anything that could come up during the install process as well as helping everyone involved become more familiar with the project.

ii. The post-sales engineer should come prepared to go through the drawings they've created and explain the logic behind the line diagrams as well as explain any complexities of the project.

iii. The AV Field Engineer should come to the meeting having already reviewed the drawing set and have any questions prepared for the meeting to clear up any confusion about the systems.

iv. The project managers should come to the meeting prepared to discuss the full deployment schedule.

e. **Notes**

i. This meeting is required before any install, bench or programming work can begin on the project.

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XIV. Client Scope Review

a. **Owner: AV Project Manager**

b. **Action Item:** Schedule a client scope review meeting with the client, AV PM and pre-sales engineer.

c. **Summary**

   i. The purpose of this meeting is to verify the full AV scope with the client so that everyone is in agreement on what they will be getting once the install is finished.

d. **Details:**

   i. This meeting can happen remote or in-person.

   ii. The internal team should bring the following documents: Scope of work, final proposal, AV drawings.

   iii. The goal of this meeting is to get verbal confirmation of all of our room scopes prior to any install work beginning (including bench work and programming).

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XIV. Programming

a. **Owner: AVSS**

b. **Action Item:** All preliminary programming for the project.

c. **Summary**

i. Programming will be required on any project which has complex DSP configuration or a customized control system (touch panels, button panels, etc). Note: programming is not automatically required on one-touch join systems if they do not have additional room controls needed in the system or a DSP that needs to be configured.

d. **Details**

i. Initial programming should always happen offsite and preferably before we mobilize to start the install.

ii. Any touch panel layouts that need to be approved by the customer should be mocked up in software and provided to the post-sales engineer to review and send to the customer for sign off before we mobilize to finish the programming.

iii. Any testing that needs to be performed prior to install should happen at the office on the AV bench or anywhere offsite that is sufficient for these tests.

iv. Once the initial offsite programming is performed and the install has taken place, the programmer will mobilize to the site for final testing/commissioning/tuning of the programming and audio systems.

e. **Notes**

i. It is imperative that ALL AV install work is completed before the programmer mobilizes to the jobsite. If programming work needs to be phased out for a larger project, please work directly with the programmer to schedule the dates of each room so they can prioritize rooms that will be completed first.

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XV. Bench Work/Staging

a. **Owner: AV Field Engineer**

b. **Action Item:** Field Engineer to stage equipment and do firmware updates/configuration of equipment.

c. **Summary**

i. Most projects will have some bench work time billed as part of the proposal. This bench work will be performed on the AV bench by the AV Field Engineers. Bench work is billed per room or room type in the proposal. Please ensure that all bench work time is accounted for when scheduling this work.

d. **Details**

i. Firmware Updates: Any firmware updates on any devices that need it should be updated prior to installing in the field.

ii. IP Addressing: All IP addresses should be preloaded into any devices that need them unless the client has specified that they prefer devices are set to DHCP. Use the AV Networking worksheet template for identifying proper IP schema with a client.

iii. Equipment Documentation: All equipment documentation should be captured in a spreadsheet before the project mobilizes for onsite install. This could include but is not limited to serial numbers, MAC addresses, IP addresses, room the equipment will be installed in, etc.

e. **Notes**

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XVI. Schedule Install

a. **Owner: AV Project Manager**

b. **Action Item:** Schedule technicians for onsite install.

c. **Summary**

   i. During the programming/bench work process, the project managers should be mapping our original install timeline into official dates. These dates should be communicated with the shared services team as soon as they are known. Any changes in schedule due to customer delays or internal delays should be communicated immediately as well so the shared services team can prioritize resources accordingly.

d. **Details**

   i. An SSP should be developed and presented which shows the exact dates each room (or potentially each part of each room) will be installed so that everyone involved knows which rooms to prioritize first.

e. **Notes**

   i. For some projects, this process will be as simple as scheduling a date or a couple of dates that we will mobilize to install a few rooms. For others, this could be a very long detailed SSP which notes exactly how we will manage a larger project. All relevant information should be communicated via teams to the entire project team.

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XVII. Onsite Install

a. **Owner: AV Field Engineer**

b. **Action Item:** Perform onsite install.

c. **Summary**

i. Following all programming/bench work completion, the AV Field Engineer, Lead techs and install techs will mobilize to do the initial install of the project onsite.

d. **Details**

i. The week leading up to the install, the project managers should be onsite regularly to ensure site conditions are ready for us to install and that all other trades are wrapping up so that all electrical, cable, and pathways are in place and ready for us.

ii. The project managers should also be regularly in contact with the furniture teams to ensure all furniture will be installed and ready for us to mount any equipment needed before we mobilize (this has been one of the biggest challenges to date with adding additional unwarranted hours to projects).

e. **Notes**

i. During the install process, the project managers will send out weekly snapshot updates to the customer and blind copy the entire project team so that everyone involved can see the progress of the install weekly.

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XVIII. Commissioning

a. **Owner: AV Field Engineer**

b. **Action Item:** Commission all AV systems – create a punchlist as you go.

c. **Summary**

i. Once the original install is done, the programmer and AV Field Engineer will re-mobilize to commission all systems. This includes final debugging/testing of any programming work, dialing in the DSP and tuning the audio in the room, and testing all room functionality.

d. **Details**

i. Programmers Work: The programmer will mobilize to debug any programming in the room, test and verify all of their code is working with the control system as well as do a final tuning of the DSP in the room (if there is one).

ii. AV Field Engineers Work: The AV Field Engineer will be onsite at the same time as the programmer and their job would be to test the overall functionality of the system. Systems to check would be all video routing, all audio routing (mics and speakers), start up/shut down of all systems, any conferencing capabilities should be tested and vetted as well as any room schedulers or additional functionality needs per the scope of work.

iii. Documentation: As each room is commissioned, a detailed document should be written up and commented regularly noting any additional items that need to be added or fixed as well as final notes showing that all systems are functional once that is the case.

e. **Notes**

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XIX. Testing Checklist

a. **Owner: AV Field Engineer**

b. **Action Item:** Finish all punch-list items.

c. **Summary**

i. Once the initial commissioning is completed, the technician team should mobilize to complete any leftover items that are commented as needing work on the commissioning documentation. This work should be prioritized as high so that we can move into the closeout process with the customer.

d. **Details**

i. Internal Checklist: Sometimes a client won't be heavily involved in the commissioning process. If this is the case, it is imperative that we have good documentation of the state of each system. In this case, our team should use our internal checklist as the measure of what work is left to be done. The checklist can be found in the project files for every job.

ii. External Checklist: Sometimes the GC/End Client will require a site walk with them through all spaces so a punch list can be put together for work left to be completed. If this is performed, please take the notes that the GC or end client has handed over regarding remaining work and upload it to the teams channel for the whole project team to see.

iii. Project Mangers: the project managers should take any checklist we have on work left to be done and assign an owner to each task as well as a date in which that task will be completed and communicate all dates to the project team and the end client.

e. **Notes**

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XX. Training

a. **Owner: AV Project Manager/Field Engineer**

b. **Action Item:** Train the customer on all AV systems

c. **Summary**

i. Once the final commissioning of the systems have taken place and all parts of the system are functional per the scope of work, the AV Project Manager and the AV Field Engineer will perform training for the End Client. Training hours are billed on a project wide basis (one labor line item on the quote). These hours can range from 2 hours (super small project) to 24+ hours (very large projects). Training is meant to be for necessary personnel only, meaning that we don't intend to train an entire company including regular users of the system. The target audience of the training would be someone who can relay this training to the company at large (like an IT manager or whoever will manage the AV system).

d. **Details**

i. Training should only be performed for each room type. Meaning if there are 5 huddle rooms that are identical, only one huddle room should need to have a training.

ii. Training should stick to the functional scope of the project. Try to avoid getting into questions about MS Teams and general organizational needs (i.e. What is the best method of scheduling a meeting in outlook). If these questions are asked, just let them know that all organizations prefer that these things happen in different ways and the person asking the question should defer to their IT team for better instructions on best practices on the program at hand.

e. **Notes**

i. It is fine to provide insight around zoom/teams/google meet or whatever else the customer is asking about as long as we preface it with the idea that most of these things can be done in multiple ways and we can't be responsible for instructing their employees to do things in a different way than the IT managers would prefer.

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XXI. Closeout/Sign Off

a. **Owner: AV Project Manager**

b. **Action Item:** Get with AVSS to develop as-builts and send off all close-out documents (owners manuals, warranties, drawings). Get formal sign-off on all systems. Close out job for final billing.

c. **Summary**

   i. Once the final commissioning and training of a system has taken place, a site walk should be conducted again with the end user to verify all system functionality and sign off on each individual room.

d. **Details**

   i. The sign off of the room indicates that all functionality is working per the scope of work and no additional work needs to be done. This process indicates the start of our 90 day workmanship warranty as well as a takeover by our service department for any additional work that needs to be done once we are signed off.

   ii. If we receive a call shortly after sign off saying a particular system isn't working, the service team should mobilize to determine if this is a workmanship issue or a service issue (client broke something or did something they shouldn't have done which broke the system after we left).

e. **Notes**

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XXII. Managed Services Transfer

a. **Owner: AV Project Manager**

b. **Action Item:** Introduce client contact to our service team and explain to them how to create a service ticket.

c. **Summary**

i. Once a project has been closed out, the customer will be introduced to our service department as their primary point of contact moving forward. Any additional calls regarding the project should be routed directly to the service department, not to the project manager.

d. **Details**

i. Once a project is officially closed and the service department has been introduced, the teams channel for this project can be officially closed as the project is completed.

ii. All service calls after this point will be documented and tracked in our service ticketing system.

iii. The service department can leverage the project team at any time if additional information or questions need to be asked.

iv. If a project warrants and entirely new quote for service work or to build additional needed functionality into an existing system, the project should be entered as a new lead and re-start from the beginning of the pre-sales process.

e. **Notes**

i. Currently the standard for AV service is either T&M without a service contract or service contracts through premiere protect. When a managed services transfer happens, we should be informing the customer how to make a service request. Service requests will be triaged by the Managed Services Manager for AVSS but then we will work with local business unit resources for deployment of technicians to resolve issues. For Premiere Protect execution, see detailed PP process.

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